Revision F18CD

Past year questions

Disclaimer

- This is just a discussion of the past year questions.
- Anything discussed here are NOT necessarily to be tested in the final exam.
- Anything NOT discussed here are NOT necessarily NOT tested in the final exam, i.e., it may be tested in the final exam.
- In short, this is just a reference, you are required to study the lecture notes, worksheets, assignments and past final exam papers, etc.

Q1. (a) Definition from Chap 1.

(b) Proof from a theorem from lecture notes, Chap 1.

(c) Prove sup/inf of a given set.

Please check whether the max/min exists or not.

If yes, then the sup/inf is immediate.

(d) Unseen question but related to max/min/sup/inf.

Most probably true/false.

Prove sup/inf:

(a) Apply Theorem 7.5 of Chap 2

(b) Apply Theorem 4.2 of Chap 1

(c) Start from the definition

Q2. (a) Definition from Chap 2.

(b) Proof from a theorem from lecture notes, Chap 2.

(c) Prove (from definition) convergence of given sequences.

(d) Unseen question but related to sequence convergence/bounded, Cauchy sequence, etc. Most probably true/false.

There are many ways we can use to find limit (or show limit does not exist) for a sequence

(a) ε - definition

(b) MCT

(c) Cauchy Criterion for sequence

(d) Subsequences tend to different limits

(e) Squeeze theorem

etc....

Q3. (a) Definition from Chap 3.

(b) Proof from a theorem from lecture notes, Chap 3.

(c) and (d) Determine/prove convergence of given series.

- must quote the test (DT, GS, PST, CT, LCT, RT, NRT, ACT, AST)
- normally alternating series will be asked
- normally will cover absolute/conditionally convergence

The difference between (c) and (d) is that (c) is a series based on (b) and (d) is normally alternating series.

For sequence that is difficult to prove that it is decreasing,

we can apply knowledge on Chap 5 (increasing/decreasing function).

One very good example iss $\sum (-1)^k \sin\left(\frac{1}{k}\right)$.

Q4. (a) Definition from Chap 4 or Chap 5.

(b) Proof from a theorem from lecture notes, Chap 4 or Chap 5.

(c) Prove limit exists/not exist at a point/on an interval; or

prove function is continuous/not continuous at a point/on an interval.

(d) Unseen, true or false. Perhaps piecewise function.